GHULAM ISHAQ KHAN INSTITUTE

of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L • Computational Methods and Tools Lab

Lab Report #7

Jacobi & Gauss-Seidel Iterative Methods

Solving Linear Systems Numerically

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Contents

Objective

The objective of this lab is to understand and implement the **Jacobi Iterative Method** for solving systems of linear equations in MATLAB. Additionally, the lab compares the Jacobi method with the **Gauss-Seidel** variant, and investigates the effect of **diagonal dominance** on convergence and divergence of these iterative solvers.

By the end of this lab, students will be able to:

- Understand the mathematical formulation of the Jacobi iterative method.
- Implement Jacobi and Gauss-Seidel methods in MATLAB.
- Verify the diagonal dominance condition and predict convergence behavior.
- Visualize iteration convergence and divergence through plots.

Theory

▶ Jacobi Iterative Method

For a linear system $A\mathbf{x} = \mathbf{b}$, the Jacobi method isolates each variable from its respective equation:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij}\, x_j^{(k)} \right)$$

All updates use values from the **previous** iteration k. The method converges when the coefficient matrix A is **strictly diagonally dominant**:

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \forall\, i$$

▶ Gauss-Seidel Method

The Gauss-Seidel method improves upon Jacobi by using **updated values immediately**:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij}\, x_j^{(k+1)} - \sum_{j>i} a_{ij}\, x_j^{(k)} \right)$$

This typically leads to **faster convergence** (fewer iterations) compared to Jacobi for the same system.

▶ Diagonal Dominance Check

lightBlue **Diagonally Dominant**	**NOT Diagonally Dominant**
$\lvert a_{ii}\rvert \geq \sum_{j\neq i} \lvert a_{ij}\rvert$ Jacobi **converges** Values settle to solution	$\lvert a_{ii}\rvert < \sum_{j\neq i} \lvert a_{ij}\rvert$ Jacobi may **diverge** Values grow unboundedly

1 Task 01 — Jacobi Method (Diagonally Dominant)

▶ Question

Implement the Jacobi Iteration Method and verify convergence for the following **diagonally dominant** system. Use initial guess $x_0 = [0,\ 0,\ 0]^T$ and $n = 20$ iterations.

$$\begin{cases} 10x + 2y + z = 9 \\ 2x + 8y + z = 13 \\ x + y + 5z = 7 \end{cases}$$

▶ Theory / Manual Analysis

Diagonal Dominance Check:

$$\begin{aligned} &\text{Row 1:} \quad |10| > |2| + |1| = 3 \quad \checkmark \\ &\text{Row 2:} \quad |8| > |2| + |1| = 3 \quad \checkmark \\ &\text{Row 3:} \quad |5| > |1| + |1| = 2 \quad \checkmark \end{aligned}$$

Strictly diagonally dominant $\Rightarrow$ **Jacobi will converge**.

Jacobi Update Equations:

$$x^{(k+1)} = \frac{9 - 2y^{(k)} - z^{(k)}}{10}, \quad y^{(k+1)} = \frac{13 - 2x^{(k)} - z^{(k)}}{8}, \quad z^{(k+1)} = \frac{7 - x^{(k)} - y^{(k)}}{5}$$

Exact solution (by elimination): $x \approx 0.5246,\ y \approx 1.3661,\ z \approx 1.0219$.

▶ MATLAB Code

```
%lab7q1 2023405

x(1) = 0;
y(1) = 0;
z(1) = 0;

n = 20;

% Jacobi iterations      use ONLY previous values
for j = 1:n
```

```
    x(j+1) = (9  - 2*y(j) - z(j)) / 10;
    y(j+1) = (13 - 2*x(j) - z(j)) / 8;
    z(j+1) = (7  - x(j)   - y(j)) / 5;
end

% Print iteration table
for j = 1:length(x)
    fprintf('Iteration %d\t: x = %.6f\t, y = %.6f\t, z = %.6f\n', ...
            j-1, x(j), y(j), z(j));
end

% Plot convergence
figure;
plot(0:n, x, '-ro', 'LineWidth', 2, 'DisplayName', 'x');
hold on;
plot(0:n, y, '-bs', 'LineWidth', 2, 'DisplayName', 'y');
plot(0:n, z, '-gd', 'LineWidth', 2, 'DisplayName', 'z');
grid on;
xlabel('Iteration');
ylabel('Value');
title('Jacobi Method Convergence');
legend('Location', 'best');
```

▶ Output

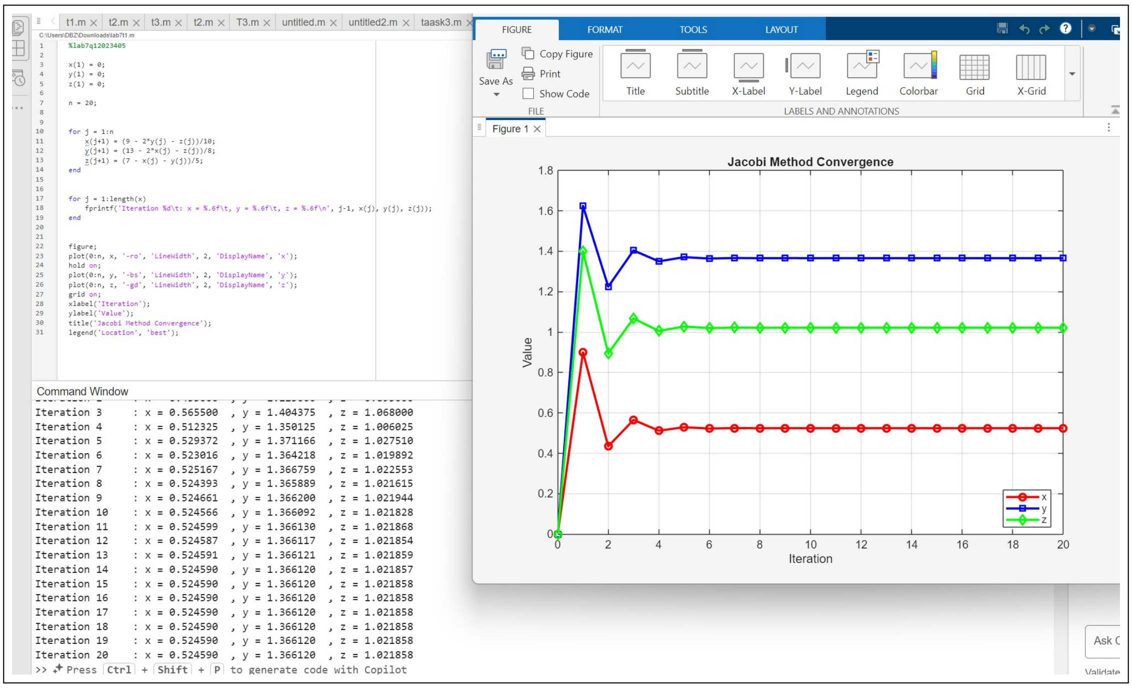

Figure 1: Task 01 — Jacobi convergence for diagonally dominant system (iterations + plot)

▶ Discussion

Converged successfully. The iteration table shows all three variables stabilizing by approximately iteration 14–15:

$$x \approx 0.524590, \quad y \approx 1.366120, \quad z \approx 1.021858$$

The convergence plot shows the characteristic oscillating decay — variables overshoot slightly in early iterations then settle. This confirms the diagonal dominance condition guarantees convergence for the Jacobi method.

2 Task 02 — Jacobi Method (Non-Diagonally Dominant)

▶ Question

Analyze the behavior of the Jacobi method on a **non-diagonally dominant** system. Use initial guess $x_0 = [0,\ 0,\ 0]^T$ and $n = 20$ iterations.

$$\begin{cases} 2x + 3y + z = 5 \\ 4x + y + 2z = 6 \\ 3x + 2y + z = 7 \end{cases}$$

▶ Theory / Manual Analysis

Diagonal Dominance Check:

$$\begin{aligned} &\text{Row 1:} \quad |2| < |3| + |1| = 4 \quad \times \ \text{(fails)} \\ &\text{Row 2:} \quad |1| < |4| + |2| = 6 \quad \times \ \text{(fails)} \\ &\text{Row 3:} \quad |1| < |3| + |2| = 5 \quad \times \ \text{(fails)} \end{aligned}$$

Not diagonally dominant $\Rightarrow$ **Jacobi is expected to diverge**.

Jacobi Update Equations:

$$x^{(k+1)} = \frac{5 - 3y^{(k)} - z^{(k)}}{2}, \quad y^{(k+1)} = \frac{6 - 4x^{(k)} - 2z^{(k)}}{1}, \quad z^{(k+1)} = \frac{7 - 3x^{(k)} - 2y^{(k)}}{1}$$

▶ MATLAB Code

```
%lab7 q2 2023405
x(1) = 0;
y(1) = 0;
z(1) = 0;

n = 20;

% Jacobi iterations     non-diagonally dominant system
for j = 1:n
    x(j+1) = (5  - 3*y(j) - z(j)) / 2;
```

```
    y(j+1) =  6  - 4*x(j) - 2*z(j);
    z(j+1) =  7  - 3*x(j) - 2*y(j);
end

% Print iteration table
for j = 1:length(x)
    fprintf('Iteration %d\t: x = %.6f\t, y = %.6f\t, z = %.6f\n', ...
             j-1, x(j), y(j), z(j));
end

% Plot     expect divergence
figure;
plot(0:n, x, '-ro', 'LineWidth', 2, 'DisplayName', 'x');
hold on;
plot(0:n, y, '-bs', 'LineWidth', 2, 'DisplayName', 'y');
plot(0:n, z, '-gd', 'LineWidth', 2, 'DisplayName', 'z');
grid on;
xlabel('Iteration');
ylabel('Value');
title('Jacobi Method (Non-Diagonally Dominant)');
legend('Location', 'best');
```

▶ Output

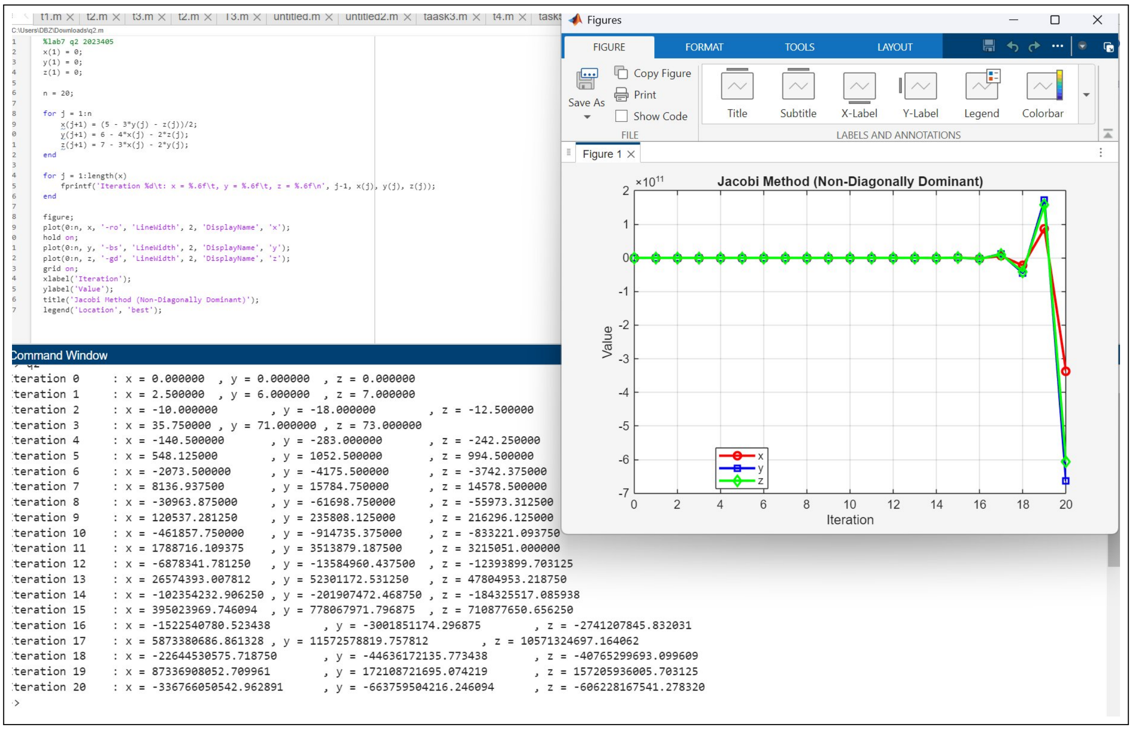

Figure 2: Task 02 — Jacobi divergence for non-diagonally dominant system

▶ Discussion

Diverged — as predicted. The iteration values grow rapidly and unboundedly. By iteration 20, values reach magnitudes of $\sim 10^{11}$. This confirms that the diagonal dominance condition is not merely a mathematical formality — violating it causes the Jacobi method to fail completely, making the output meaningless for root-finding purposes.

Conclusion

This lab demonstrated iterative methods for solving 3×3 linear systems in MATLAB:

- **Task 1 (Jacobi — Convergent):** The diagonally dominant system converged to $(x, y, z) \approx (0.5246, 1.3661, 1.0219)$ after approximately 14 iterations. The convergence plot showed oscillating decay toward the fixed point, confirming the guarantee provided by diagonal dominance.

- **Task 2 (Jacobi — Divergent):** The non-diagonally dominant system diverged explosively, with values reaching $\sim 10^{11}$ by iteration 20. All three rows failed the diagonal dominance check, and the resulting plot confirmed unbounded growth, making the method completely unreliable for this system.

Key Takeaway: Always verify diagonal dominance before applying the Jacobi method. When $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all rows, convergence is guaranteed. Failing this condition risks divergence and meaningless results.